# Nirmit Hitendra Dagli

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## PROFESSIONAL SUMMARY

Applied AI and AI infrastructure engineer with 5+ years taking systems from first conversation to production. Founding engineer at Sidecoach Sports: built a Claude agent suite (MCP, RAG, golden-set evals, fine-tuning) that cuts alert noise ~75% for the engineering team. Benchmarks GPU inference (vLLM vs. SGLang: 609 vs. 384 tokens/s at 64 users; speculative decoding, continuous batching, KV cache, parallelism) and ran 50+ Kubernetes clusters in production. Got enterprise RAG approved inside a regulated bank; 7 years of teaching. CKA, CKAD, AWS Solutions Architect; MS Computer Science; 5 IEEE and Springer papers.

## WORK EXPERIENCE

**Founding Platform Engineer | Sidecoach Sports | Waco, TX** **Aug 2026 – Present**

- **AI Agent Suite (Claude):** Designed and shipped a production suite of Claude-powered agents on Azure AI Foundry, used by the engineering team, connected to Azure, Jira, and Teams through a custom MCP agent service, with human approval gates before agents act.
- **Alert Triage & RCA:** Agents cut alert noise ~75%: of every 20+ alerts, they flag the ~5 real incidents, open Jira tickets, run approved restarts, draft root-cause analyses, and surface recurring patterns; also produce cost reports and security checks.
- **RAG:** Built the RAG pipeline that grounds agent answers and proposed fixes in internal docs and past incidents, tuning chunking and retrieval against the golden set.
- **Evals, Golden Sets & Fine-Tuning:** Built a golden set from live sources (Kaggle datasets, live vulnerability feeds, and real alert events) and an automated eval harness run on prompt, model, and retrieval changes; fine-tuned a model on curated domain examples and measured it against the base model before rollout.
- **AI Gateway & Compliance Agent:** Built the company AI gateway (Entra ID OIDC, per-team API keys, PII redaction, OpenTelemetry tracing for usage, latency, and cost) and an AI agent that audits the website for compliance.
- **Platform, Security & Compliance:** First infrastructure, security, and AI hire; built the Azure platform, GitHub CI/CD, and dev, staging, and prod from scratch, cutting monthly Azure spend 50%; raised Microsoft CSPM Secure Score from 25% to 100%; own COPPA, FERPA, and PCI DSS (SAQ-A) compliance and SOC 2 readiness.

**Research Assistant | Quinnipiac University | Hamden, CT** **Sep 2024 – May 2026**

- **GPU Infrastructure & LLM Serving:** Ran Kubernetes GPU clusters (NVIDIA GPU Operator, Cilium) and Slurm GPU job scheduling, serving Llama 3, Mistral, Gemma, and DeepSeek with 4-bit quantization behind a process-isolated FastAPI server that handles model loading and swapping across GPUs and eliminated CUDA out-of-memory crashes.
- **Inference Optimization:** Applied speculative decoding in vLLM and SGLang, prefill/decode disaggregation, and tensor and pipeline parallelism with NCCL across the lab's multi-GPU nodes to serve open-source LLMs.
- **Small Language Model:** Built and trained a small GPT-style transformer language model from scratch as a graduate research project for the MS Generative AI course.
- **Inference Engineering & Benchmarking:** Benchmarked vLLM, SGLang, and TensorRT-LLM on the lab's GPUs for throughput, time-to-first-token, and GPU memory; compared on-prem GPUs, GCP L4, and AWS Bedrock for a RAG workload (30s → 15s → 10s avg latency), leading the lab to move inference to Bedrock.
- **Applied AI Research:** Built SPARK, a plug-and-play RAG framework (IEEE UEMCON 2025), and HERO / HERO-S workload-aware schedulers (SEET and IEEE ISAIA 2026) routing AI jobs across CPU, GPU, cloud, and edge (8×H100 cost model), cutting energy 72–98% vs. cloud-only.

**Cloud and AI Engineer Intern | State Street | Boston, MA** **Jun 2025 – Aug 2025**

- **Internal AI Assistant:** Shipped an internal AI chatbot: Confluence RAG on Azure AI Foundry, delivered as a Microsoft Teams bot that answers access questions and helps developers write Terraform modules; secured approvals from the Confluence, Jira, Azure, and network teams.
- **Amazon Q Rollout:** Rolled out Amazon Q across 44 AWS accounts under bank-grade security with the Security, Compliance, and AWS enablement teams; trained engineers on Copilot and AI-assisted coding.

**Senior Infrastructure Engineer | Zycus | Mumbai, India** **Jan 2021 – Jun 2024**

- **AI & Automation:** Built the company's first internal AI chatbot on Azure OpenAI (2024); automated recurring service requests through Jenkins pipelines, cutting tickets ~80% (about 50 to under 10 per week).
- **Kubernetes & Multi-Cloud Platform:** Led the migration of 2,000+ workloads into 50+ Kubernetes clusters (EKS, AKS, GKE, Rancher, OpenShift) with ArgoCD GitOps; led a team of 5 standardizing Terraform, Ansible, and Jenkins provisioning, cutting provisioning time ~40%.

- **Ransomware Recovery:** First responder in the company-wide recovery; restored 1,500+ servers across 7 environments (4 production) in 15 days and led 4 months of security hardening.
- **Networking & Security:** Designed the multi-account AWS network (Transit Gateway, Site-to-Site VPN, Direct Connect) and rolled out Palo Alto, Zscaler, CrowdStrike, and CyberArk company-wide.

## PROJECTS

**llm-serving-bench | Open-source LLM serving benchmark (GitHub)** **Oct 2026**

- **vLLM vs. SGLang:** Built a harness that runs both engines on the same GPU, model, prompts, and 85% GPU memory (KV cache) budget, with unique prompts so prefix caching can't skew results; across 2,048 requests on a Tesla T4, vLLM delivered 609 vs. 384 tokens/s at 64 users and 1.8s vs. 7.4s median TTFT, and ruled out attention kernels (both on Triton) as the cause, pointing to scheduling and batching.
- **Continuous Batching & Quantization:** Showed continuous batching raised vLLM throughput 12x from 1 to 64 users (51 to 609 tokens/s) while per-user decode slowed from 20 to 92 ms/token; AWQ 4-bit made single-user decode 2.3x faster but added only 5% at 64 users, since decode is memory-bandwidth bound.

**GPU Inference Platform on Kubernetes** **Oct 2026 – Present**

- **LLM Serving Stack:** Building a vLLM serving platform on Kubernetes (NVIDIA GPU Operator) with autoscaling on request queue depth, DCGM, Prometheus, and Grafana GPU dashboards, and a GPU failure runbook (XID errors, ECC, drain and replace).

## TECHNICAL SKILLS

**GenAI & Agents:** Claude, Azure AI Foundry, Azure OpenAI, AWS Bedrock, Amazon Q, Hugging Face, LangChain, MCP, RAG, embeddings and vector search, prompt and context engineering, fine-tuning, evals (golden sets, regression testing), guardrails, PII redaction

**Inference & GPU:** vLLM, SGLang, TensorRT-LLM, speculative decoding, prefill/decode disaggregation, tensor and pipeline parallelism, continuous batching, KV cache (PagedAttention, prefix caching), AWQ and 4-bit quantization, model loading and swapping, workload scheduling, throughput, TTFT, and TPOT benchmarking, NVIDIA GPU Operator, DCGM, NCCL, Slurm, FastAPI, GCP L4, AWS Bedrock

**Languages & Tools:** Python, Bash, Swift, Git, GitHub Actions, Jira

**Cloud:** Azure (AKS, Entra ID, Intune, Defender, Monitor); AWS (EC2, EKS, S3, IAM, CloudFront); GCP (GKE, Vertex AI, Google AI Studio)

**Kubernetes & IaC:** Kubernetes (CKA, CKAD), Docker, Helm, Terraform, Bicep, Ansible, ArgoCD

**Observability & Security:** OpenTelemetry, Prometheus, Grafana, ELK, Dynatrace, Zero Trust, CrowdStrike, CyberArk, SOC 2, PCI DSS, FERPA, COPPA

## EDUCATION

**Quinnipiac University | Master of Science in Computer Science | GPA: 3.9/4.0** **May 2026**
**University of Mumbai | Bachelor of Engineering in Information Technology** **May 2020**

## PUBLICATIONS & CERTIFICATIONS

- **Hybrid AI Scheduling & IoT Security:** HERO-S, **IEEE ISAIA** **2026**
- **Hybrid AI Scheduling:** HERO, **Springer SEET** **2026**
- **Sustainable AI:** Energy-Efficient AI for Foundation Models (coauthor), **Computers (MDPI)** **2026**
- **Generative AI:** SPARK – Smart Plug-and-Play AI Framework for RAG & Knowledge, **IEEE UEMCON** **Oct 2025**
- **Quantum Computing:** Grover's Algorithm for String Search and Its Practical Limits, **Springer SEET** **Aug 2025**
- LangChain for LLM Application Development, **DeepLearning.AI** **Jun 2025**
- Certified Kubernetes Application Developer – **CKAD** **Jan 2025**
- Certified Kubernetes Administrator – **CKA** **Mar 2024**
- AWS Certified Solutions Architect – Associate, **Amazon Web Services** **Nov 2022**